Prodigy Anesthesia Quick Review

Anticholinergics/Cholinergic Agonists

What are the three anticholinergic drugs in current use for anesthesia?	Atropine, scopolamine, and glycopyrrolate *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 267.*
With what receptors do anticholinergic drugs combine?	They combine reversibly with the muscarinic cholinergic receptors and prevent acetylcholine from binding to the receptor. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 266.*
Will the administration of atropine or glycopyrrolate to a parturient increase the heart rate of the fetus?	Although atropine can cross the placenta (glycopyrrolate cannot), there is no significant change in fetal heart rate after intravenous administration to the mother. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 269.*
How do anticholinergics affect airway resistance?	The bronchodilatory effects of anticholinergics reduce airway resistance and increase anatomic dead space. The effect is more pronounced in patients with asthma and COPD. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 235.*
Which of the three anticholinergics has the greatest sedative effect?	Scopolamine has the greatest sedative effect. Atropine causes slight sedation and glycopyrrolate has no sedative effect at all. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 268.*
How do the anticholinergic drugs compare with respect to their antisialagogue effect?	Scopolamine has the greatest antisialagogue effect, followed closely by glycopyrrolate, then atropine. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 234.*
How does the anticholinergics compare with respect to their tendency to increase heart rate?	Atropine causes the greatest increase in heart rate, followed by glycopyrrolate, then scopolamine. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 268.*
Can the anticholinergic drugs be administered orally?	Even the lipid-soluble anticholinergics, atropine and scopolamine, are not absorbed with enough predictability to warrant oral administration. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 268.*

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How does the onset of an increased heart rate from administration of IV atropine compare with glycopyrrolate?	Atropine will produce an increase in the heart rate in about 1 minute. Glycopyrrolate takes about 2-3 minutes to increase the heart rate. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 268.*
Which anticholinergic is considered the safest for use in patients with glaucoma? Which is the least safe?	Scopolamine exhibits the greatest mydriatic effect and requires the greatest amount of caution in patients with glaucoma. Atropine 0.4 mg to 1 mg IV is considered safe when administered with an anticholinesterase drug because little to no increase in pupil size is noted. Glycopyrrolate has little to no mydriatic effect. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 268.*
How does the effect of atropine on the heart rate differ between infants, young adults, and the elderly?	Vagal tone is enhanced in young adults, and the increase in heart rate is most evident in this population. In infants, and in the elderly, even large doses of atropine may have little effect on the heart rate. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 270.*
How do anticholinergics affect gastric function?	Administration of anticholinergics in general results in a decrease in gastric secretions, decreased peristalsis and intestinal motility, prolonged gastric emptying time, and reduced lower esophageal sphincter tone. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 235.*
What is ipratropium?	Ipratropium is an inhaled anticholinergic administered to produce bronchodilation in the treatment of asthma. It is a derivative of atropine that has limited systemic absorption from the respiratory tract. It is used (often in combination with albuterol) in the treatment of chronic obstructive pulmonary disease. It exerts no change on heart rate or intraocular pressure. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 271-272.*
What are the indications for transdermal scopolamine?	Transdermal scopolamine is useful in preventing nausea in patients with a history of motion sickness. It also does not have the sedative, cycloplegic, or tachycardic side effects of parenteral scopolamine. It should be administered 4 hours prior to the stimulus for the greatest effect. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 273.*

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How do anticholinergics affect gastric pH?	Anticholinergics increase gastric pH and have been used in the treatment of peptic ulcer disease. None of them are selective for this, however, and large doses are required to affect gastric pH. This results in a high incidence of undesirable side effects in many patients. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 271-272.*
Other than increased heart rate, what effect do anticholinergic drugs have on the electrocardiogram?	They shorten the PR interval *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 270.*
What is central anticholinergic syndrome?	Scopolamine and atropine both cross the blood-brain barrier and block muscarinic cholinergic receptors in the CNS, producing restlessness, hallucinations, somnolence, and potentially, unconsciousness. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 273.*
What is the treatment for central anticholinergic syndrome?	The anticholinesterase physostigmine administered at a dose of 15-60 mcg/kg IV will treat the symptoms of central anticholinergic syndrome. Physostigmine is a lipid-soluble tertiary amine that is able to cross the blood-brain barrier. Edrophonium, neostigmine, and pyridostigmine are quaternary ammoniums that do not cross the blood-brain barrier easily and are not appropriate treatments for this syndrome. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 76.*
What anticholinergic has been used as a treatment for hiccups?	Atropine 0.5 mg IV has been reported as an effective treatment for hiccups that occur after the placement of a laryngeal mask airway. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 273.*
What drugs can predispose a patient to central anticholinergic syndrome?	Tricyclic antidepressants (like amitryptiline), antipsychotics, and antihistamines have antimuscarinic characteristics which can potentiate anticholinergics and predispose the patient to central anticholinergic syndrome. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 236-237.*

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